

UNMANNED AERIAL VEHICLE

EMD51802 DESIGN PROJECT II

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HINDUSTAN

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BONAFIDE CERTIFICATE

Certified that this project report titled "**UNMANNED AERIAL VEHICLE**" is the bonafide work of **Shaik Sameer Salahuddin (22125004), Thirisha. R (22125022), Deepak (22125060)** who did the task under my supervision during the academic year 2023 – 2024.

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ABSTRACT

Unmanned Aerial Vehicles (UAVs) represent a transformative technology with vast potential in safety and agricultural domains. This project focuses on Quadcopters, a subtype of UAVs characterized by four rotors, aiming to address challenges related to wireless control effectiveness and design optimization. The research begins with an overview of UAVs, emphasizing their significance in modern aviation as autonomous or remotely piloted aircraft. Quadcopters, being a popular choice due to their maneuverability and versatility, are the primary focus of this study.

One of the main challenges identified is the lack of effective control mechanisms for Quadcopters during wireless operation. This limitation hampers their utility in critical applications such as safety operations and agricultural surveillance. Through comprehensive analysis and experimentation, the project seeks to identify innovative solutions to enhance wireless control reliability and responsiveness. The construction and working principles of Quadcopter drones are thoroughly investigated, with a focus on understanding the CC3D microcontroller, a key component in drone navigation and control systems. By delving into the intricacies of the CC3D microcontroller, the project aims to develop insights into its programming and functionality, enabling more efficient control algorithms and flight performance.

Ultimately, the project aims to contribute to the development of Quadcopter UAVs for various applications by leveraging advancements in wireless control systems, design optimization, and microcontroller technology. Through practical experimentation and iterative design processes, the project seeks to produce innovative solutions that enhance the reliability, versatility, and safety of Quadcopter UAVs in diverse operational environments.

Keywords: Unmanned Aerial Vehicles, wireless control systems, Quadcopter drones

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DEPARTMENT OF MECHATRONICS ENGINEERING

EMD51802 DESIGN PROJECT II

Project Batch No : 17

Title of The Project : Unmanned Aerial Vehicle

Number of The Students In The Batch : 3

Student's Individual Contribution

Name of the Student: Shaik Sameer Salahuddin

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Contribution content is reported on pages: Hardware intergration(17) and functional testing (19)

I have gone through the research papers published previously and came up with the idea to Integrate the components. I analyzed the working mechanism and helped in the hardware integration.

Shaik Sameer Salahuddin (22125004)



DEPARTMENT OF MECHATRONICS ENGINEERING

EMD51802 DESIGN PROJECT II

Project Batch No : 17

Title of the project : Unmanned aerial vehicle

Number of the students in the batch : 3

Student's Individual Contribution

Name of the Student: Thirisha. R

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I have gone through the research papers published previously and came up with the idea to design the Unmanned aerial vehicle. I analyzed the working mechanism and helped in the functional testing.

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CHAPTER -1

INTRODUCTION

Unmanned Aerial Vehicles (UAVs), particularly Quadcopters, have emerged as transformative assets, revolutionizing industries with their unparalleled capabilities and versatility. This paper offers an in-depth exploration of Quadcopter technology, spanning design considerations, operational principles, technological integration, industry applications, challenges, future directions, and societal implications.

Quadcopter design encompasses intricate considerations to optimize performance and efficiency. Aerodynamic refinement ensures stability and maneuverability, while structural enhancements guarantee durability under various conditions. Payload capacity optimization strikes a balance between performance and operational requirements. Moreover, advancements in materials science and manufacturing techniques enable the development of lightweight yet robust structures, enhancing flight endurance and agility. Case studies exemplify innovative design approaches, such as foldable or modular designs, and their impact on Quadcopter performance and usability in different environments.

The operational prowess of Quadcopters relies on sophisticated flight control algorithms, sensor systems, and communication protocols. Flight control algorithms govern stabilization, navigation, and autonomy, enabling precise control in dynamic environments and under varying payloads. Sensor systems, including gyroscopes, accelerometers, GPS, and LiDAR, provide crucial environmental perception and situational awareness, enhancing operational effectiveness and safety. Communication protocols facilitate seamless interaction between the Quadcopter and ground stations, ensuring reliable data transmission and command execution. Additionally, advancements in software-defined radio (SDR) technology enable adaptive communication strategies, enhancing resilience in congested or noisy RF environments.

Integration of advanced technologies such as microcontrollers, wireless control systems, and artificial intelligence enhances Quadcopter capabilities. Microcontrollers, exemplified by the CC3D and Pixhawk, serve as the brain of the Quadcopter, orchestrating flight control algorithms, sensor data processing, and payload management. Advancements in wireless control systems, including frequency hopping spread spectrum (FHSS) and digital transmission protocols, enable robust communication and telemetry, even in adverse conditions. Integration of artificial intelligence techniques, such as machine learning and computer vision, fosters

autonomy, adaptive behavior, and intelligent decision-making, enabling Quadcopters to navigate complex environments and execute missions with minimal human intervention.

Quadcopters find application across a spectrum of industries, including safety and emergency response, agriculture, infrastructure inspection, construction, and entertainment. In safety and emergency response, Quadcopters facilitate search and rescue operations, disaster management, and environmental monitoring by providing real-time situational awareness and access to inaccessible or hazardous areas. In agriculture, they revolutionize crop monitoring, pest control, and precision agriculture practices, enabling farmers to optimize resource allocation, increase yield, and reduce environmental impact. Infrastructure inspection benefits from Quadcopters' ability to access hard-to-reach areas, such as bridges, power lines, and pipelines, while minimizing downtime and operational costs. In construction, Quadcopters provide aerial surveying, progress monitoring, and site mapping services, improving project management efficiency and accuracy. Furthermore, entertainment industries utilize Quadcopters for aerial photography, cinematography, and live event coverage, offering breathtaking aerial perspectives and immersive experiences to audiences worldwide.

Despite their potential, Quadcopters face regulatory challenges, technological limitations, and operational constraints that require careful consideration. Regulatory frameworks must evolve to accommodate UAV integration into airspace systems while addressing privacy, security, and safety concerns. Technological challenges include power efficiency, scalability, robustness, and interoperability, necessitating ongoing research and development efforts in areas such as battery technology, propulsion systems, and sensor fusion algorithms. Moreover, human factors, such as pilot training, human-machine interaction, and workload management, play a critical role in ensuring safe and effective Quadcopter operations. Future directions involve advancements in autonomy, swarming capabilities, and urban air mobility, paving the way for a connected and efficient future where Quadcopters seamlessly integrate into daily life and critical infrastructure. Quadcopter UAVs epitomize innovation and versatility, reshaping industries and driving technological progress. By addressing design considerations, operational principles, technological integration, industry applications, challenges, and future directions, this paper highlights the transformative potential of Quadcopters.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

This collection of papers and theses represents a comprehensive exploration into the expansive domain of Unmanned Aerial Vehicle (UAV) technology and its wide-ranging applications across diverse fields. Each contribution offers a unique perspective and insight, collectively enriching our understanding of the capabilities and potential of UAVs in modern society.

Beginning with the proposal for an autonomous aerial surveyor, the research sets forth a vision for leveraging cutting-edge technologies to equip UAVs with advanced functionalities such as aerial surveillance, object recognition, tracking, and 3D mapping. This initiative aims to address the needs of both civil and military sectors, offering a cost-effective solution that could revolutionize operations in the burgeoning UAV market.

Subsequent investigations delve into the realm of construction management, where UAVs have emerged as invaluable tools for enhancing efficiency and safety. Through a meticulous review of existing literature, these studies categorize and analyse findings related to UAV applications in construction, shedding light on methodologies, challenges, and prospects. The insights gleaned from this review are poised to inform and guide future research and development efforts in this vital area. Moreover, the evaluation of UAVs for Structural Health Monitoring (SHM) in civil infrastructure underscores their potential to revolutionize maintenance and inspection practices.

By detailing applications, categorizing UAV types, and proposing innovative neural network-based damage detection models, researchers pave the way for more accurate and efficient performance assessment across various computing platforms. In parallel, the role of UAVs in civil applications, particularly precision agriculture, emerges as a focal point of exploration. Highlighting their cost-effectiveness and capacity for integration with artificial intelligence (AI), these studies underscore the transformative impact of UAV technology on agricultural practices, promising increased productivity and sustainability in the sector.

2.2 INFERENCE FROM RESEARCH PAPERS

Saurav Kumar, E.Kanniga, Literature Survey On Unmanned Aerial Vehicles, International Journal of Pure and Applied Mathematics, Volume 119 No. 12 2018, 4381-4387.[1]

Inference: The proposal aims to pioneer the development of an autonomous aerial surveyor, harnessing modern technologies to revolutionize aerial surveillance, object recognition, tracking, and simple 3D mapping. By leveraging the advancements in UAV technology, such as AI, computer vision, and sensor integration, the autonomous aerial surveyor seeks to offer a cost-effective solution catering to both civil and military applications in the burgeoning UAV market. Through autonomous operation, the aerial surveyor promises increased efficiency and accuracy in data collection, enabling timely and informed decision-making. Its capabilities in aerial surveillance and object recognition hold significant potential for enhancing security measures, disaster response, and environmental monitoring. Furthermore, by providing simple 3D mapping capabilities, the aerial surveyor opens avenues for applications in urban planning, infrastructure development, and land management. With its versatility and cost-effectiveness, the autonomous aerial surveyor emerges as a transformative solution poised to meet the evolving needs of diverse industries and sectors in the UAV landscape.

Andres Acero Molina , Yilei Huang , Yuhan Jiang, A Review of Unmanned Aerial Vehicle Applications in Construction Management, 2016–2021. Standards 2023, 3, 95–109.[2]

Inference: Unmanned Aerial Vehicles (UAVs), or drones, have emerged as powerful tools in construction management, offering capabilities ranging from site surveying and mapping to construction monitoring and inspection. This review categorizes UAV research in construction management into key areas such as site surveying, monitoring, project management, and environmental monitoring, highlighting findings and implications for future applications. Key findings include improved efficiency, safety, and decision-making capabilities, while future directions point toward advancements in UAV technology, integration with emerging technologies like AI and machine learning, and collaboration to address regulatory and privacy challenges. With continued innovation and collaboration, UAVs hold the potential to significantly enhance efficiency, safety, and sustainability in the civil and construction industries.

Dr. Wael Zatar, Dr. Haroon Malik Dr. Jamil M Chaudri, A Systematic Literature Survey Of Unmanned Aerial Vehicle-Based Structural Health Monitoring, 2020.[3]

Inference: The thesis provides a comprehensive evaluation of Unmanned Aerial Vehicles (UAVs) for Structural Health Monitoring (SHM) in civil infrastructure, categorizing UAV types, elucidating their diverse applications, and proposing neural network-based damage detection models for robust performance assessment across computing platforms. Through meticulous categorization, the thesis delineates the varied capabilities of UAVs, from fixed-wing platforms suitable for large-scale inspections to agile quadcopters adept at maneuvering in confined spaces. It details the wide-ranging applications of UAVs in SHM, including visual inspections, thermal imaging, and LiDAR scanning, highlighting their efficacy in detecting structural defects and assessing overall conditions. Furthermore, the thesis introduces neural network-based models tailored for damage detection, leveraging the power of machine learning to analyze UAV-acquired data and accurately identify structural anomalies. By emphasizing the scalability and adaptability of these models across computing platforms, from onboard UAV systems to cloud-based processing, the thesis underscores their potential to revolutionize SHM practices and enhance infrastructure safety and resilience.

Mithra Sivakumar, Naga Malleswari TY, A Literature Survey of Unmanned Aerial Vehicle Usage for Civil Applications, v13, e4021, 2021.[4]

Inference: The paper underscores the expanding significance of Unmanned Aerial Vehicles (UAVs) in civil applications, particularly in precision agriculture, by accentuating their cost-effectiveness and potential for augmented functionality through integration with artificial intelligence (AI). It elucidates how UAVs have become indispensable tools in modern agriculture, offering unparalleled capabilities for crop monitoring, pest management, and resource optimization. By leveraging AI algorithms, UAVs can process vast amounts of aerial data to provide actionable insights, ranging from identifying crop health issues to optimizing irrigation schedules. This integration not only enhances the efficiency and accuracy of agricultural operations but also enables farmers to make informed decisions that drive productivity and sustainability. As UAV technology continues to evolve and AI integration becomes more prevalent, the paper underscores the transformative potential of this synergy in revolutionizing agricultural practices and addressing the challenges of feeding a growing global population.

A.M. Reasad Azim Bappy , MD. Asfak-Ur-Rafi ,MD. Saddamul Islam ,Ali Sajjad , Khan Nafis Imran , Design and Development of Unmanned Aerial Vehicles (Drone) for Civil Applications 2022.[5]

Inference: The research presents a groundbreaking versatile drone system equipped with RF control, MATLAB simulation capabilities, and Android-GPS integration, poised to revolutionize a wide array of applications spanning security, disaster management, and surveillance. By combining RF control for precise maneuverability, MATLAB simulation for comprehensive system analysis and optimization, and Android-GPS integration for seamless navigation and real-time data acquisition, this drone system offers unparalleled versatility and adaptability. In security applications, it can be deployed for perimeter surveillance, intruder detection, and crowd monitoring, enhancing situational awareness and response capabilities. In disaster management scenarios, the drone system facilitates rapid aerial reconnaissance, damage assessment, and search and rescue operations, aiding first responders in critical decision-making processes. Furthermore, its surveillance capabilities extend to environmental monitoring, wildlife conservation, and infrastructure inspection, contributing to enhanced safety, efficiency, and sustainability across various domains.

2.3 SUMMARY OF THE RESEARCH PAPER

The proposal not only recognizes the transformative potential of UAV technology but also delves deeper into its applications across diverse industries, showcasing its versatility and adaptability to meet evolving needs. By introducing an autonomous aerial surveyor tailored for aerial surveillance, object recognition, and 3D mapping, the paper addresses the increasing demand for efficient and cost-effective solutions in both civil and military sectors. Furthermore, it extends the discussion to the realm of construction management, where UAVs play a crucial role in Structural Health Monitoring (SHM). The proposal advocates for the adoption of neural network-based damage detection models, highlighting their potential to revolutionize structural assessment practices and enhance infrastructure safety. Additionally, the paper sheds light on UAVs' significance in precision agriculture, offering insights into crop monitoring, pest management, and resource optimization. This application underscores UAVs' capacity to drive sustainability and productivity in the agricultural sector while minimizing environmental impact. Moreover, the introduction of a versatile drone system capable of serving diverse applications such as security and disaster management underscores the multifaceted nature of

UAV technology. By showcasing the breadth of UAV applications, the proposal underscores its growing significance as a disruptive force across industries, paving the way for innovation, efficiency, and economic growth on a global scale.

2.3 RESEARCH GAP

Research gaps in drone technology include enhancing autonomy, improving sense and avoid capabilities, increasing energy efficiency, addressing security and privacy concerns, refining regulatory frameworks, establishing standardization, mitigating environmental impact, understanding human-drone interaction, optimizing data processing, and exploring integration with other technologies.

CHAPTER 3

METHODOLOGY

3. 1 Selection of Components (Hardware/Software)

S.No	COMPONENTS	QUANTITY
1	1000kv brushless motor	4
2	CC3D flight controller	1
3	4200mah 3s lipo battery and charger	1
4	30A ESC(electronic speed controllers)	4
5	10x4.4 Inches propellers	4
6	Flysky FS i6 Transmitter and receiver	1
7	F450 Quadcopter frame	1

Table 1. List of components

1000kv brushless motor

- Motor KV: 1000
- Current Capacity: 12A/ 60S
- LiPO Batteries: 2S-3S
- Max. Efficiency Current: 4-10A
- Shaft Diameter: 3.17 mm



Fig 1 1000kv brushless motor

A 1000-kilovolt brushless motor rotates at 1000 revolutions per minute for each volt applied. Four 1000-kilovolt brushless motors can elevate a combined weight of 3.2–4 kilograms. It possesses effectiveness, requires minimal upkeep, and finds application in RC vehicles, drones, and electric aircraft. Its dimensions, mass, voltage, and current specifications differ based on the model and intended use.

Figure 1 describes 1000kv brushless motor which has three pins signal(yellow),ground(Black) and VCC(Red)

CC3D flight controller

As shown in figure 2 The CC3D stands out as a favoured flight controller for multirotor aircraft. Equipped with gyroscopes and accelerometers, it ensures stability during flight and grants users the flexibility to adjust flight settings using software such as LibrePilot or LibrePilotor.



Fig 2 CC3D controller



Fig 3 software

As shown in figure 3 LibrePilot emerges as an open-source autopilot platform crafted for managing unmanned aerial vehicles (UAVs) and other robotic systems. The program boasts compatibility with a broad array of hardware components, encompassing flight controllers, GPS modules, and telemetry systems, empowering users to construct and tailor their drones to precise specifications. Featuring an intuitive interface, LibrePilot facilitates the utilization of advanced functionalities such as waypoint navigation, altitude retention, and stabilization modes. Furthermore, it extends comprehensive support for diverse flight modes, comprising manual, stabilized, acro, and autonomous modes, catering to the needs of both novices and seasoned operators. Leveraging its open-source ethos, LibrePilot fosters collaboration and ingenuity within the UAV community, inviting users to actively participate in its evolution and tailor it to meet their unique demands.

Pin configuration

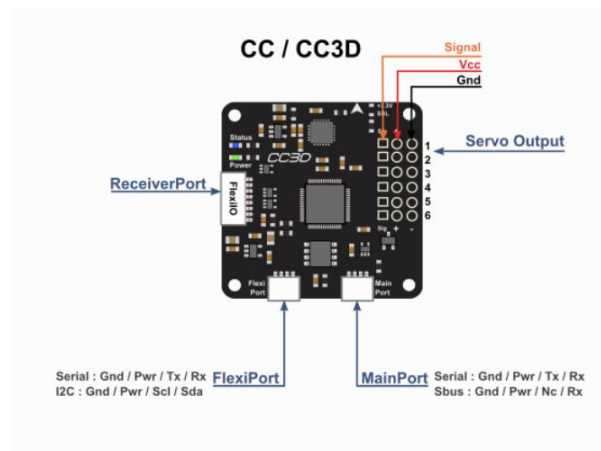


Fig 4 CC3D pinouts

Figure 4 describes In this setup, the receiver port and servo output are utilized.

Servo Outputs 1-6: These channels deliver PWM signals to servos or Electronic Speed Controllers (ESCs). Generally, power is supplied through these connectors from one of the ESCs. The diagram and the board indicate the positive (Vcc) and negative (Gnd) pins.

The layout of the servo output pins is as follows:

Outer Pin → Ground

Middle Pin → 5V - 15V

Inner Pin → Signa

Receiver Port Pins

Channel 1	THROTTLE	White	Signal 1
Channel 2	AILERON or ROLL	Blue	Signal 2
Channel 3	ELEV or PITCH	Yellow	Signal 3
Channel 4	RUDDER	Green	Signal 4
Channel 5	GEAR - Flight mode	Orange	Signal 5
Channel 6	AUX1	Purple	Signal 6

Table 2 receiver port pins

4200mAh 3s lipo battery and charger

Specification :

- Capacity (mAh) :420
- Weight (g) 319
- Output Voltage (VDC): 11.1
- Charge Rate (C): 1 ~ 3
- Length (mm): 140
- Width (mm): 43
- Height (mm): 25



Fig 5 4200mAh 3s lipobattery

With a 4200mAh 3s LiPo battery, a drone may typically fly for approximately 15 to 30 minutes, depending on its size, weight, and power consumption. Flight time can vary based on factors such as wind conditions, altitude, and flight.

Figure 5 describes that a lipo battery of 4200mah has inlet and outlet ports to charge and function the xt60 port.

30A ESC (ELECTROIC SPEED CONTROLLLER)

Specification:

- Constant Current: 30A (Max 40A<10s).
- BEC: 5V 3A.
- Input Voltage: 11.1 ~ 11.7 V.
- Li-Po battery: 2S ~ 3S.
- Ni-MH 4-10 CELLS Auto Detect

As shown in figure 6 30A ESC (Electronic Speed Controller) regulates motor speed in remote-controlled vehicles, handling up to 30 amps of continuous current, crucial for precise control and efficient operation in UAVs and RC vehicles.



Fig 6 30A ESC

10*4.4. inches propellers.

These propellers are 10 inches in diameter with a pitch of 4.4 inches. They generate lift to propel the quadcopter and are typically made of lightweight materials like plastic or carbon fiber.



Fig 7 propellers

Figure 7 has two propellers and the adjustment washers

Flysky Fs i6 Transmitter and receiver

The Flysky FS i6 stands out as a widely-used radio transmitter and receiver system employed for managing RC vehicles, including drones. It functions within the 2.4GHz spectrum and ensures dependable command over the quadcopter.

Specification:

Table 3

Channels	6
RF range	2.4055 - 2.475 GHz
Bandwidth	500KHz
Band	140
RF Power	Below 20 dBm
A system with 2.4G	AFHDS 2A & AFHDS 2B
Sensitivity	1024
Low voltage warning	Below 4.2V
DSC Port	PS2
Output	PPM
Charger Point	No
Weight	392gm
Power	6V 1.5AA*4
Display mode	Type positive transfective STN type, 128*64 dot matrix
Size	174*89*190mm

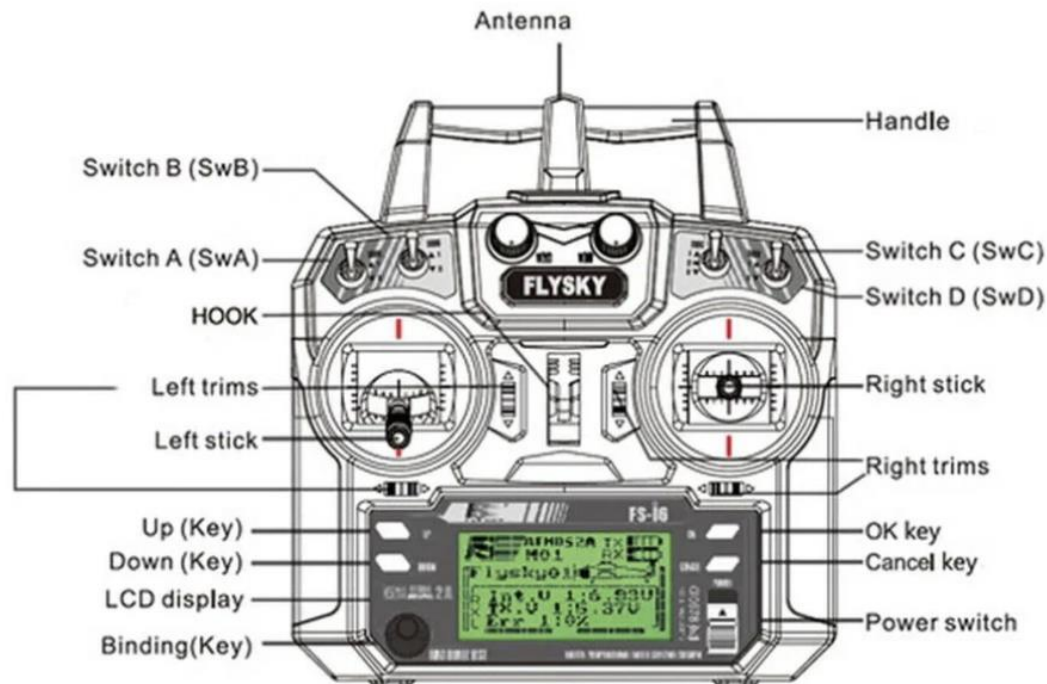


Fig 8 Fly sky i6 transmitter

The primary benefit of this transmitter lies in its independence from a PC or laptop for configuration. Equipped with an LCD display and user-friendly buttons, setup is effortlessly managed directly on the device. It boasts an operational range of up to 1500 meters, albeit subject to magnetic interference. Higher levels of interference result in reduced range, whereas lower interference levels allow for extended operation distances.

Figure 8 describes

Display Panel: Serving as the interface of this transmitter, the display panel facilitates user interaction. Utilizing the buttons, users navigate through the LCD interface, where various indicators such as battery status and mode are visible. The battery indicator communicates the power level and operational status of both the transmitter and receiver. It's worth noting that the receiver's battery status remains invisible unless it's bound to the transmitter.

Control Sticks: This transmitter features a pair of control sticks, positioned on the left and right sides respectively. These control sticks, also known as gimbals, enable manipulation of four channels: Channel1 - Ailerons (Roll), Channel2 - Elevator (Pitch), Channel3 - Throttle, and Channel4 - Rudder (Yaw); these settings represent the default configuration for mode 2.

Further clarification on the mode is provided below. In the accompanying image, two control sticks are clearly depicted, one on each side.

Trim Controls: Trims serve the purpose of stabilizing the drone during flight. When the drone experiences destabilization, these trims are adjusted accordingly. Each of the four channels is equipped with its respective trim control. By holding the trim in the desired direction, users can expedite trim adjustments. When the trim position reaches the center, the transmitter emits a higher-pitched beeping tone.

Rotary Controls: Positioned atop the transmitter, these two knobs serve as auxiliary channels. They offer the flexibility to attach switches to additional channels, enabling control over various components of the model such as landing gear and headlights. In the provided image, two rotary controls are visible, with the left one designated as VRA and the right one as VRB.

Toggle Controls: Positioned atop this transmitter are four switches: toggle A (SWA), toggle B (SWB), toggle C (SWC), and toggle D (SWD). SWA, SWB, and SWD are dual-mode switches, while SWC is a triple-mode switch. Channels 5 and 6 functions can be assigned to either of the dual-mode switches. SWC is utilized when GPS functionality is employed in drones.

Binding Button: The binding button is employed during the process of synchronizing the transmitter with the receiver. Positioned at the bottom of the left side of the transmitter, the binding button, depicted in black, facilitates this synchronization.

Receiver



Fig 9 FS-IA6 Receiver

This receiver features two antennas and supports six channels. To ensure optimal signal quality, it is recommended to mount the receiver away from motors or metallic components.

Connectors are utilized to establish connections between various components of the model and the receiver.

Figure 9 shows there are 7 channels ch1 to ch6 is for servo functions and ch7 is for vcc.

Channels 1 through 6: These channels serve as connection points for the ESC (Electronic Speed Controller), VCC, and other components. B/VCC: This port facilitates the connection of the binding cable to synchronize the receiver with the transmitter.

F450 Quadcopter frame

The F450 Quadcopter Frame (figure 10) is made using 35% Glass Fiber, making it sturdy and durable. Its arms are manufactured from highly resilient polyamide nylon, presenting reinforced moulded arms with considerable thickness, minimizing the risk of arm breakage at motor mounts during harsh landings. Additionally, these arms feature support ridges, which improve stability and facilitate faster forward flight. As shown in figure 11 The frame offers durability, lightweight construction, and a functional layout, integrating a Printed Circuit Board (PCB) that enables ESCs to be directly soldered onto the Quadcopter.

Specification

- Wheelbase: 450mm
- Material: Glass Fiber + Polyamide-Nylon
- Motor Mounting Hole Diameter: 3 mm
- Arm Size: 220 x 40 (LxW) mm
- Arm mounting holes (on frame): 3 mm
- Arm mounting holes (on arm): 2 mm



Fig10 Drone Frame

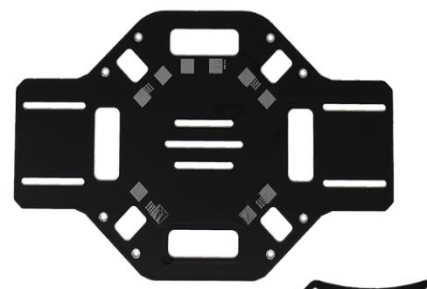


Fig 11 power distribution board

3.2 BLOCK DIAGRAM AND EXPLANATION

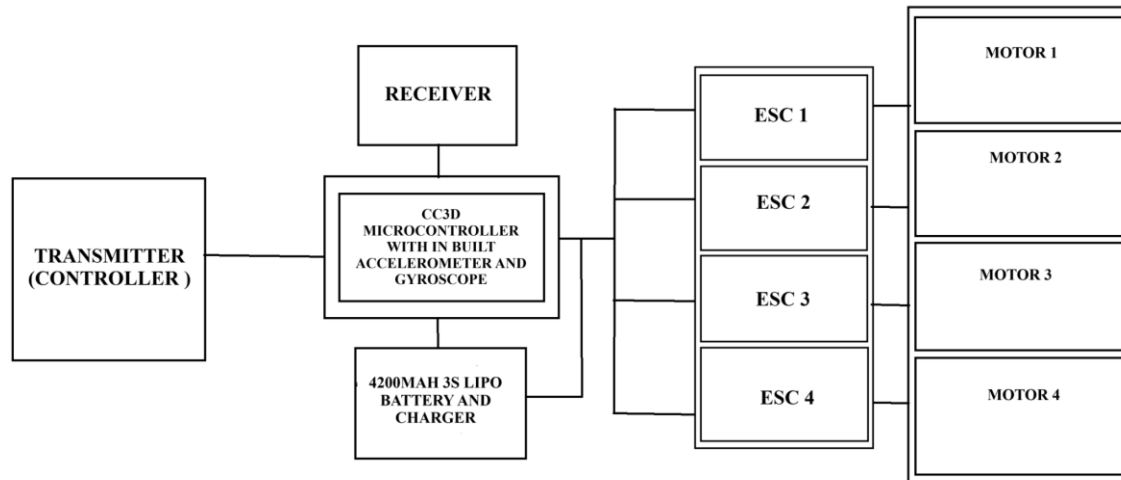


Fig 12 Block diagram

According to Figure 12 The CC3D flight controller serves as the central hub for communication and control within the drone system, receiving power directly from the battery and commands from the receiver or autonomous flight software. Linked to each of the four Electronic Speed Controllers (ESCs), the flight controller regulates the flow of electricity to the 1000kV brushless motors, dictating their speed and rotation to maintain precise control and stabilization of the drone during flight. The LiPo battery supplies the necessary electrical energy to power the ESCs and flight controller, while the Flysky FS i6 transmitter and receiver system enable remote control over the drone's movements and functions, ensuring seamless operation and responsiveness. This interconnected system allows for efficient and reliable flight performance, empowering the drone to execute maneuvers with precision and agility across various applications.

3.3 HARDWARE INTERGRATION

3.3.1 Building Frame



Fig 13 Building Frame



Fig 14 Frame with stand

The F450 Quadcopter Frame is used , first bottom plate and arm of the drone is assembled. Stand is been added for lading purpose.(figure 13 and 14)

3.3.2 Soldering ESC

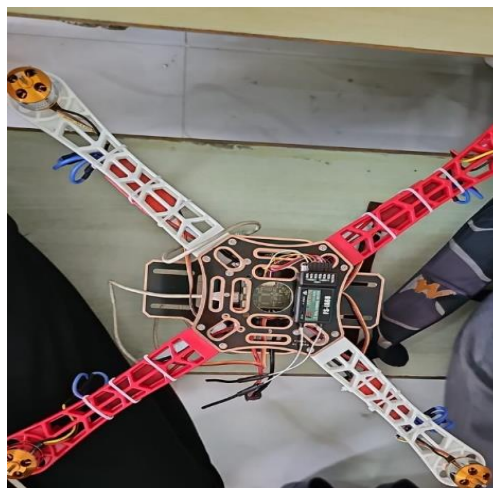


Fig 15 Soldering ESC

Soldering ESC(figure 15) to the inbuilt power distribution board, the motors are fixed in the motor mount and ESC is connected with the motors. After soldering top plate is been fixed to mount the CC3D microcontroller and Flysky Fs i6 receiver.

3.3.3 Final assembly

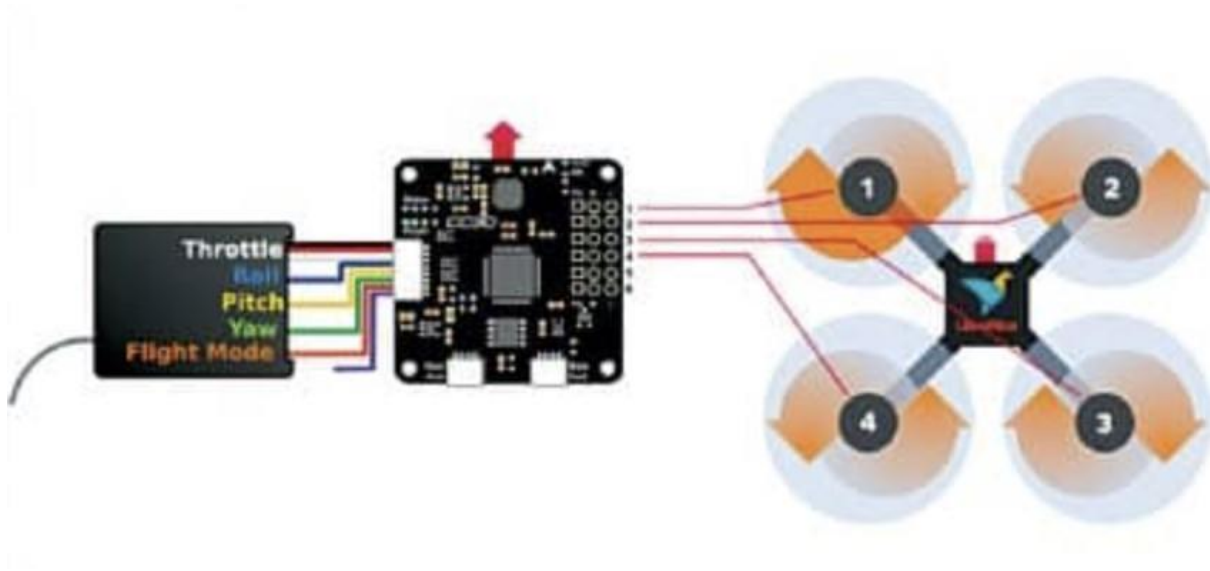


Fig 16 connection diagram

According to the figure 16 , the CC3D receiver pins are connected to the Flysky Fs i6 receiver such as throttle,roll, pitch, and yaw. All the servo output pins are connected to ESC such as VCC, GND, Signal.



Fig 17 final assembly

CC3D is placed in the center of the top plate, for calibration of the drone. To the side receiver is placed. All components such as the brushless motor, battery, ESC, and propellers are assembled (figure 17)

3.4 WORKING

3.4.1 Interfacing CC3D with LibrePilot

Linking the CC3D flight controller with LibrePilot entails establishing a connection between the CC3D board and a computer equipped with the LibrePilot GCS (Ground Control Station) software using a USB cable. After establishing this connection, individuals can customize and calibrate the CC3D through the LibrePilot GCS, modifying parameters like PID values, flight modes, and receiver configuration to fine-tune the drone's performance according to their preferences and needs.

To interface the CC3D with LibrePilot, first, ensure the drone's battery is disconnected. Then, mount the CC3D onto the drone frame and place it on a level surface for calibration. This ensures accurate sensor calibration and optimal performance during flight. Once the CC3D is securely mounted and leveled, connect it to a computer running LibrePilot to configure its settings and calibrate sensors as needed for smooth operation.



Fig 18 Assembled components



Fig 19 wizard setup

After selecting the firmware and configuration, initiate the firmware upgrade process. This ensures that the CC3D is equipped with the latest firmware and configured properly for quadcopter operation.



Fig 20 upgrade

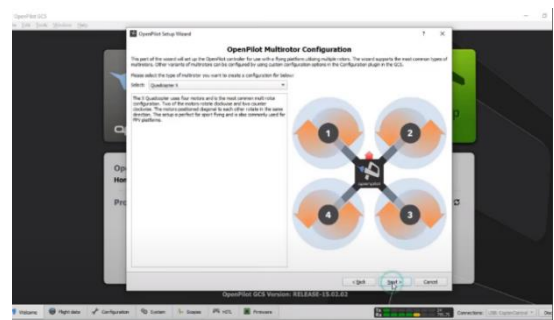


Fig 21 Model selection

3.4.2 ESC calibration procedure

1. Confirm all safety procedures,
2. Press the start button when it is enabled, the battery is not connected,
3. Connect the battery to your airframe,
4. Wait for ESC calibration beep(s),
5. Press the stop button,
6. Wait for ESC confirmation beep(s),
7. Do the same for all ESC
8. Disconnect the battery

After selecting the "Calculate" button, the software begins the process of calibrating the sensors. Once this calibration process is complete, then proceed to connect the battery to the drone. This sequence ensures that the sensors are accurately calibrated before powering up the drone with the battery.



Fig 22 ESC Calibration



Fig 23 calculating

After connecting the battery, each motor is calibrated individually. This ensures that each motor is synchronized and operating correctly before the drone takes flight.

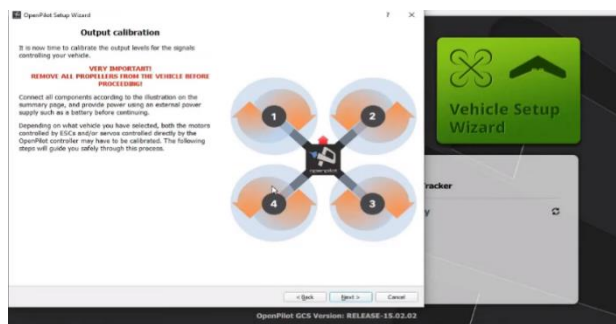


Fig 24 battery connection

3.4.3 Interfacing Flysky FS i6 Transmitter and Receiver

To interface the transmitter and receiver for drone control, first, ensure they are paired and powered on. Then, configure the transmitter settings to assign channels for pitch, throttle, yaw, and roll control. This setup allows for precise control over the drone's movements in flight.

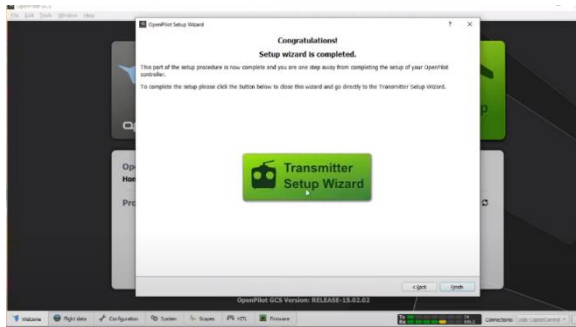


Fig 25 Transmitter setup

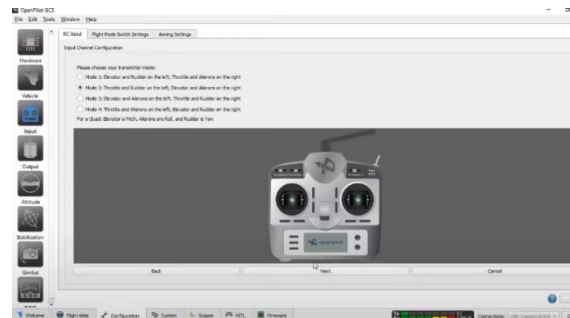


Fig 26 Assigning Channels

The joysticks are utilized to maneuver the aircraft, with each joystick serving two functions. The right joystick governs pitch and roll, while the left joystick manages throttle and yaw.

Pitch Adjustment: Commonly referred to as elevator operation, in Mode 2, pitch control is allocated to Channel 2 on the right-side joystick.

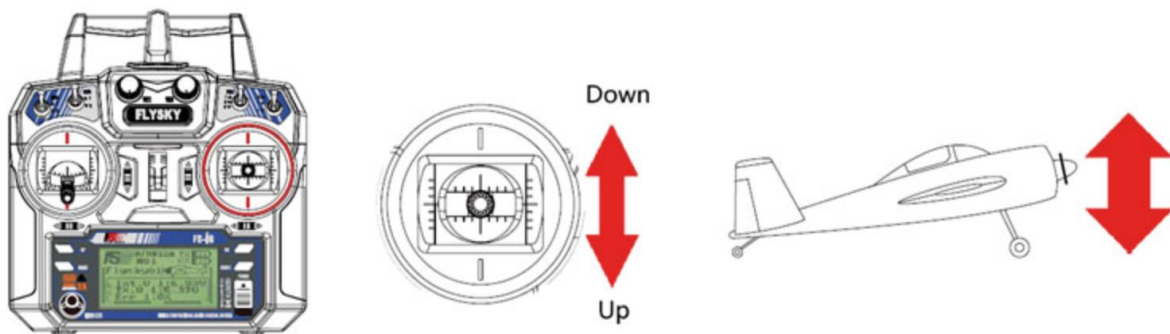


Fig 27 pitch

Roll Adjustment: Known as aileron operation, in Mode 2, roll control is assigned to Channel 1 on the right-side joystick

Yaw Adjustment: Also referred to as rudder operation, in Mode 2, yaw control is allocated to Channel 4 on the left-side joystick.

Throttle Adjustment: In Mode 2, throttle control is managed in Channel 3 on the left-side joystick

Roll (Right Stick Left/Right)

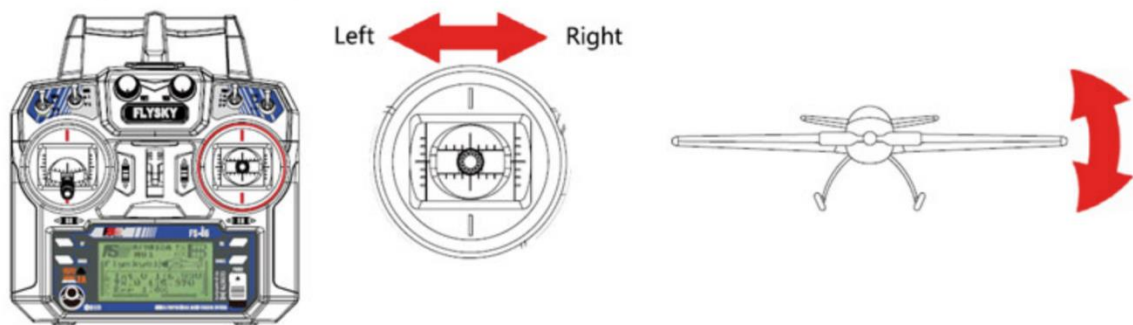


Fig 28 Roll

Yaw (Left Stick Left/Right)

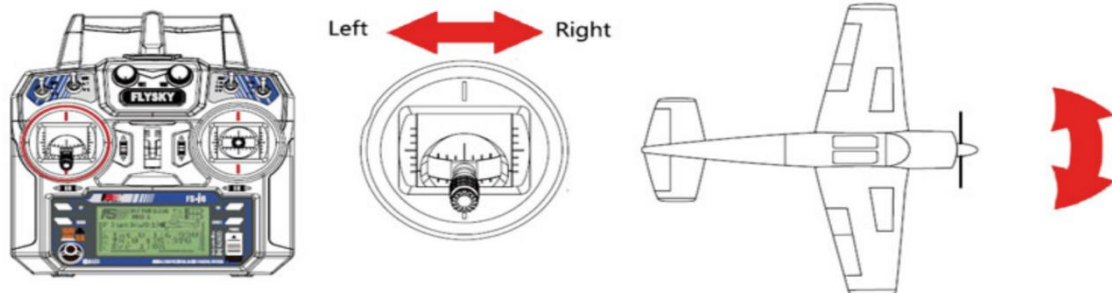


Fig 29 Yaw

Throttle (Left Stick Up/Down)

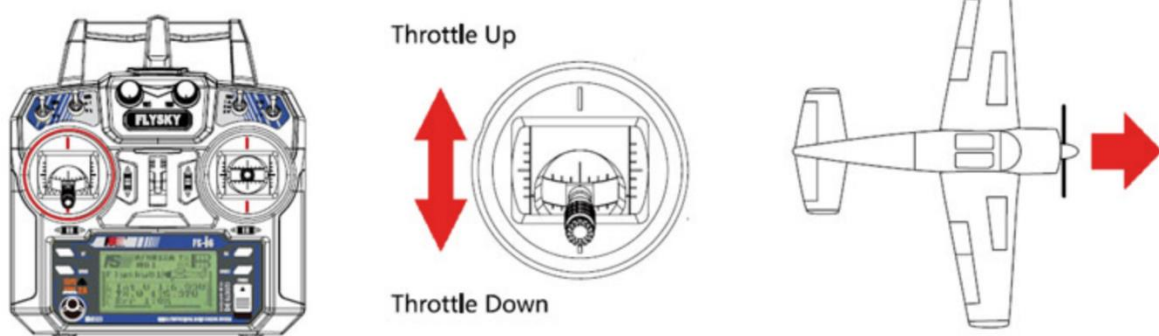


Fig 30 Throttle

CHAPTER -4

RESULT



Fig 31 Fabricated model

CHAPTER -5

CONCLUSION

In conclusion, the rise of Quadcopter UAVs stands as a testament to the remarkable advancements in unmanned aerial technology, ushering in a new era of versatility, efficiency, and innovation across diverse industries and applications. With their unparalleled agility, stability, and adaptability, Quadcopters have become indispensable assets in fields ranging from safety and disaster management to agriculture and environmental monitoring. By offering essential capabilities that enhance operational efficiency, effectiveness, and safety, these aerial vehicles have revolutionized traditional approaches to various tasks, empowering organizations and individuals to achieve unprecedented levels of productivity and success.

Moreover, the ongoing evolution of Quadcopter technology, driven by a commitment to innovation and the integration of cutting-edge advancements like the CC3D microcontroller, continues to push the boundaries of performance and reliability. Through meticulous design approaches, utilization of advanced materials, and exploration of novel system integration techniques, projects aimed at enhancing Quadcopter platforms are laying the groundwork for transformative breakthroughs in wireless control systems and design strategies. These initiatives not only seek to improve the capabilities of Quadcopter UAVs but also foster a deeper understanding of their operational principles, paving the way for further innovation and advancement in the field. Looking forward, the continued evolution and widespread adoption of Quadcopter UAVs hold tremendous potential for reshaping industries and domains worldwide. By enabling data-driven decision-making, enhancing operational capabilities, and improving overall efficiency, Quadcopters are poised to drive significant advancements in infrastructure management, precision agriculture, emergency response, and beyond. As these technologies mature and expand their reach, they will unlock new opportunities for exploration, discovery, and problem-solving, fueling a new wave of innovation and progress in unmanned aerial technology.

In conclusion, the journey of Quadcopter UAVs represents a testament to human ingenuity, innovation, and perseverance. From their humble beginnings as experimental prototypes to their current status as indispensable tools in modern society, Quadcopters have continuously pushed the boundaries of what is possible in unmanned aerial technology.

With their unrivaled agility, stability, and adaptability, these aerial vehicles have transformed industries, empowered individuals, and reshaped our understanding of the world. As we embark on the next phase of their evolution, guided by curiosity, collaboration, and a commitment to excellence, the potential for Quadcopters to revolutionize even more aspects of our lives is limitless. Through continued innovation, research, and exploration, we can unlock new opportunities, overcome challenges, and build a brighter future with Quadcopter UAVs leading the way.

CHAPTER 6

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